MEASURED ENERGY CURRICULUM · v4.0

Measured Energy

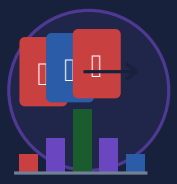
Professional Teacher Guide

Teacher-facing edition · planning, safety, timing, facilitation, and assessment

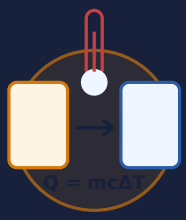
MKS · Statistical Mechanics · Thermodynamics · Purple Power · Fair Fitness Testing



MKS
ruler · balance · clock



Stat Mech
cards · counts · patterns



Thermo
heat · ΔT · equilibrium



Fair Power
sport drink · steps · equity

When many small events interact, group behavior becomes measurable, predictable, and useful.

UMA Motion LLC · May 2026

Table of Contents

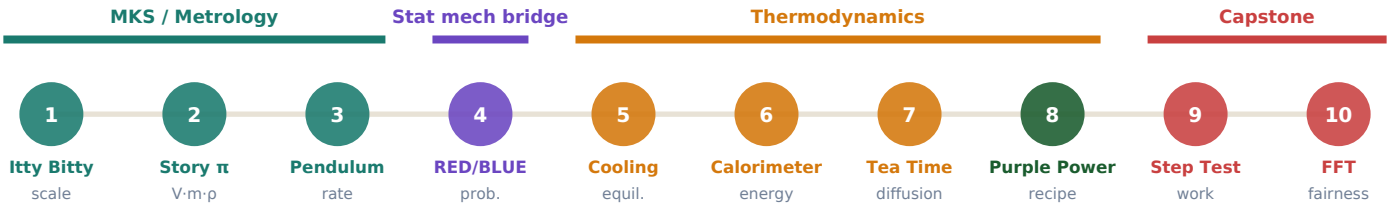
Executive Overview 3
Curriculum Spine 4
Safety, Equity, and Assessment 5
Implementation Guide 6
Lab 1: Itty Bitty PLUS 7
Lab 2: Story of π / DIY Graduated Cylinder 8
Lab 3: Seconds Pendulum / Water Clock 9
Lab 4: RED/BLUE Card Mixing 10
Lab 5: Newton’s Cooling / Predictive Mixing 11
Lab 6: Pop-Can Calorimeter / Conservation 12
Lab 7: Tea Time: Organic Extraction 13
Lab 8: Purple Power: Design Your Own Sport Drink 14
Lab 9: Harvard Step Test 15
Lab 10: FFT: Fair Fitness Testing for Populations 16
Final Synthesis & Extension 17
Teacher Glossary 18

Executive Overview

Measured Energy is a ten-lab curriculum arc that begins with the MKS language of measurement, introduces statistical mechanics through a visible RED/BLUE card activity, develops core thermodynamic reasoning, and ends with Purple Power and Fair Fitness Testing as humane capstone applications.

Designed for grades 8–10; Labs 1–4 may be used as standalone middle-school introductions, while Labs 5–10 extend naturally into high-school physical science, biology, health science, and introductory engineering. Advanced classes can add spreadsheet modeling, curve fitting, and efficiency/error analysis.

Governing sentence: When many small events interact, group behavior becomes measurable, predictable, and useful.



The curriculum works because each lab inherits a habit from the prior lab: make a measurement, notice a pattern, use a model, then decide what the model can and cannot claim. The RED/BLUE activity is not a decorative probability detour. It is the conceptual bridge that makes entropy and population fairness intuitive before equations become heavy.

CURRICULUM CLUSTERS

Cluster	Labs	Essential idea
MKS / Metrology	1–3	Length, mass, time, volume, rate, and accumulation become reliable tools.
Statistical Mechanics Bridge	4	Many exact arrangements produce predictable broad patterns.
Thermodynamics	5–7	Heat, energy, diffusion, and equilibrium are measured through familiar systems.
Capstone Applications	8–10	Mixture design, human work, power, recovery, and fairness are handled with measurement ethics.

Safety, Equity, and Assessment

This curriculum deliberately avoids ranking students, bodies, or tastes. It teaches measurement as responsibility: numbers can clarify a system, but they must be collected safely and interpreted humanely.

CLASSROOM SAFETY AND RESPECT PROTOCOL

- All human-performance activities are opt-in; observer, timer, recorder, and calculator roles are legitimate participation.
- Use coded data sets for body-related work. No public ranking of body size, speed, recovery, or appearance.
- Flame, hot water, food-safe tasting, and allergy-related procedures require explicit teacher control.
- Students may critique a measurement method, but not a person.
- Every claim should state what was measured, what was inferred, and what uncertainty remains.

ASSESSMENT RUBRIC

Dimension	Beginning	Developing	Proficient	Advanced
Measurement	Records numbers without units.	Uses some units but misses controls.	Uses appropriate units and repeatable method.	Identifies uncertainty and improves method.
Model use	Copies formula.	Substitutes values with errors.	Uses model correctly with units.	Explains model limits and assumptions.
Reasoning	Gives unsupported opinion.	Links data to claim weakly.	Uses evidence to support claim.	Compares alternatives and uncertainty.
Safety / equity	Needs reminders.	Follows basic rules.	Uses respectful, safe interpretation.	Actively protects privacy and fairness.

Implementation Guide

A practical pacing plan is shown below. The full sequence can run as a two- to four-week unit depending on whether optional extensions and data-analysis sessions are included.

Lab	Suggested time	Planning note
1. Itty Bitty PLUS	~45 min	MKS / Metrology
2. Story of π / DIY Graduated Cylinder	~45–60 min	MKS / Metrology
3. Seconds Pendulum / Water Clock	1–2 class periods	MKS / Metrology
4. RED/BLUE Card Mixing	~45 min	Statistical Mechanics Bridge
5. Newton’s Cooling / Predictive Mixing	1 class period	Thermodynamics
6. Pop-Can Calorimeter / Conservation	1–2 class periods	Thermodynamics
7. Tea Time: Organic Extraction	~45–60 min	Thermodynamics
8. Purple Power: Design Your Own Sport Drink	2 class periods or one extended block	Capstone Bridge
9. Harvard Step Test	1–2 class periods	Capstone Application
10. FFT: Fair Fitness Testing for Populations	1 class period plus optional project time	Capstone Application

Teacher workflow: Keep the first pass concrete and fast. Return to mathematical depth after students have seen the phenomenon. For example, in RED/BLUE, have students shuffle and count before introducing microstate language.

Lab 1 · Itty Bitty PLUS

Scale, unit choice, repeated halving, and orders of magnitude.



scale · unit · ratio

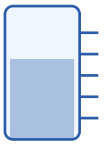
Suggested time	~45 min
Key model	scale factor = model size / real size
Materials	Paper strips or grid paper, scissors, ruler, index cards, optional magnifier, optional digital scale.
Safety	Use scissors carefully; keep small paper pieces off the floor.

Launch	Hold up a full sheet of paper and ask: “How many times can we cut this in half before it becomes hard to measure?” Then ask students what changes: the paper, the unit, or our ability to measure it.
Investigation	Students repeatedly divide a visible object, record the size after each division, and compare direct measurement with inferred size. Have them build a table of cut number, fraction remaining, estimated length, and measurable confidence.
Discussion hinge	The hinge is that measurement is not simply looking. Once objects become too small, students must infer from a pattern. This previews all later labs: measure what can be seen, then reason to what cannot be seen directly.
Assessment evidence	Look for correct unit labels, a defensible halving table, and a short explanation of why inference becomes necessary. Strong answers mention precision, scale, and limits of instruments.
Teacher emphasis	Make “unit choice” explicit. A good unit makes a hard measurement easier; a bad unit hides the pattern.

Differentiation / extension: Support: provide pre-marked strips. Extend: graph size vs. cut number on a log scale.

Lab 2 · Story of π / DIY Graduated Cylinder

Length becomes area; area becomes volume; water connects volume to mass.



1 g = 1 mL = 1 cm³

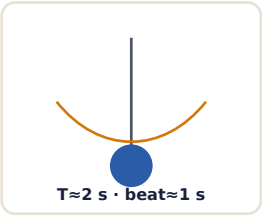
Suggested time	~45–60 min
Key model	$V = \pi r^2 h$; $\rho = m/V$; for water, 1 mL ≈ 1 cm³ ≈ 1 g
Materials	Clear cylinder or bottle, ruler, tape, water, cup, scale, marker, graph paper.
Safety	Clean spills immediately; avoid glass containers when possible.

Launch	Show a cylinder and ask: “Can we make a measuring instrument using only length?” Students should see that radius and height are length measurements, yet together they create volume.
Investigation	Students mark height intervals on a cylinder, calculate predicted volume using $\pi r^2 h$, add water, and compare measured mass with predicted volume. The scale becomes the check on the geometry.
Discussion hinge	The hinge is dimensional building: length → area → volume → mass. Water acts as the bridge because one cubic centimeter of water has a mass close to one gram.
Assessment evidence	Students should produce a calibrated cylinder scale, a calculation using $\pi r^2 h$, and a density statement that uses units correctly.
Teacher emphasis	This is not “memorize density.” It is “build density from measurements.”

Differentiation / extension: Support: use a known cylinder diameter. Extend: compare cylinder calculation with measured mass at several heights and compute percent error.

Lab 3 · Seconds Pendulum / Water Clock

Periodic motion, elapsed time, flow rate, and accumulation.



Suggested time	1–2 class periods
Key model	$T = 2\pi\sqrt{(L/g)}$; rate = Δamount/Δt; accumulated amount = rate × time
Materials	String, washers or bob, meter stick, stopwatch, cup or bottle with small hole, water, tray, scale or graduated cylinder.
Safety	Keep water away from electronics and walking paths; secure the pendulum support.

Launch	Start with a hanging bob near one meter long. Ask: “Can a swinging object become a clock?” Then ask students what must stay consistent for the swing to measure time.
Investigation	Students time 10 complete cycles at several lengths, calculate period T, and then tune the length until the half-period beat is close to 1 second (full cycle T≈2 s). In the water-clock portion they measure accumulated water mass or volume over known time intervals.
Discussion hinge	The hinge is that repeating motion and steady flow both turn time into something countable. A pendulum counts cycles; a water clock accumulates mass. Both make elapsed time measurable.
Assessment evidence	Students should identify the near-meter length of a seconds pendulum, calculate rate from accumulated water, and explain why measuring several cycles is better than timing one swing.
Teacher emphasis	Historical sidebar: Galileo made pendulum motion worth studying; Huygens made it into a precise clock. A seconds pendulum has full period about 2 seconds, with one-second beats. Eratosthenes showed the same spirit of inference: simple measurements can reveal large hidden quantities.

Differentiation / extension: Support: time 10 cycles together as a class. Extend: estimate g from measured T and L, then compare with 9.8 m/s².

Lab 4 · RED/BLUE Card Mixing

Microstates, macrostates, probability, and the intuition behind entropy.



microstates → probability

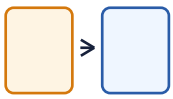
Suggested time	~45 min
Key model	number of arrangements = C(n,k)
Materials	Red and blue cards or two colored decks, recording sheet, optional histogram board.
Safety	No special hazard; keep cards organized for repeated trials.

Launch	Put all red cards together and all blue cards together. Ask: “After shuffling, why do we almost never return to this neat arrangement?” The point is not luck; it is counting.
Investigation	Students deal repeated hands, record the red/blue count, and compare exact arrangements with broad macrostates. Then they list or calculate how many arrangements belong to each count.
Discussion hinge	The hinge is the difference between a specific arrangement and a broad category. Exact order is hard to predict, but the distribution of outcomes is predictable because some categories have many more arrangements.
Assessment evidence	Students should use the words microstate and macrostate correctly and explain why mixed states are more likely without saying “because things like disorder.”
Teacher emphasis	This is the doorway into entropy. Do not rush it. The future heat labs will make more sense if students first own the counting argument.

Differentiation / extension: Support: use 5-card examples before full decks. Extend: calculate binomial probabilities or simulate 100 trials.

Lab 5 · Newton’s Cooling / Predictive Mixing

Thermal equilibrium, cooling curves, and weighted-average prediction.



ΔT shrinks over time

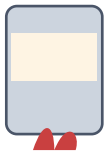
Suggested time	1 class period
Key model	$T_f = (m_{hot}T_{hot} + m_{cold}T_{cold})/(m_{hot} + m_{cold})$
Materials	Warm water, cool water, insulated cups, thermometer or temperature probe, scale or graduated cylinder, timer.
Safety	Use warm water only; avoid burns; dry spills promptly.

Launch	Ask students to predict the final temperature when a small amount of hot water meets a larger amount of cool water. Have them commit to a number before mixing.
Investigation	Students measure masses and starting temperatures, predict the final mixed temperature, mix, and compare. They then collect a cooling curve for a warm cup approaching room temperature.
Discussion hinge	The hinge is equilibrium. Predictive mixing is a “where does it end?” problem; Newton cooling is a “how does it approach the end?” problem. Both show energy spreading toward a shared condition.
Assessment evidence	Look for a correctly weighted prediction, a labeled cooling graph, and an explanation of why temperature difference drives the rate of cooling.
Teacher emphasis	Link explicitly back to RED/BLUE: many invisible molecular exchanges produce a smooth, measurable thermal pattern.

Differentiation / extension: Support: use equal masses first. Extend: fit a cooling curve or compare insulated vs. uninsulated cups.

Lab 6 · Pop-Can Calorimeter / Conservation

Energy accounting, mass change, water heating, and the First Law.



Q = mcΔT

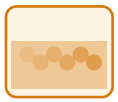
Suggested time	1–2 class periods
Key model	Q = mcΔT
Materials	Aluminum can, water, thermometer, scale, candle/food sample as approved, stand, heat-safe surface, goggles.
Safety	Teacher-supervised flame only. Tie back hair, wear eye protection, and keep water nearby. Do not eat lab samples.

Launch	Ask: “If fuel loses mass and water gains temperature, can we use the water as an energy meter?” Students should see the can as a crude but real calorimeter.
Investigation	Students measure water mass, starting temperature, final temperature, and fuel mass change. They calculate Q absorbed by water and compare energy per gram of fuel or sample.
Discussion hinge	The hinge is honest accounting: some energy heats the water, but much escapes to the air, can, and room. The First Law is not “perfect capture”; it is conservation across all pathways.
Assessment evidence	Students should calculate Q using correct units and identify at least three sources of energy loss or uncertainty.
Teacher emphasis	Use conservation of mass as a discipline check: record what changed, how much changed, and where the energy likely went.

Differentiation / extension: Support: provide calculation template for Q = mcΔT. Extend: estimate efficiency by comparing food label energy to captured water energy.

Lab 7 · Tea Time: Organic Extraction

Diffusion, extraction, color concentration, and approach to equilibrium.



diffusion · extraction

Suggested time	~45–60 min
Key model	extraction rate ≈ Δcolor/Δt; concentration changes over time
Materials	Tea bags or safe plant-based infusion, hot/warm water, clear cups, timer, thermometer, optional color scale or phone camera.
Safety	Use warm water when students handle cups; avoid allergen ingredients; tasting is optional and only with food-safe materials.

Launch	Drop a tea bag into clear water and ask: “Does the color move because the water is stirred, or because molecules are spreading?” Let students watch the first minute silently before explaining.
Investigation	Students vary steeping time, water temperature, or stirring and compare color intensity. They record temperature, time, qualitative color, and optional image-based color values.
Discussion hinge	The hinge is molecular spreading. In RED/BLUE, cards spread into many possible arrangements; in Tea Time, molecules spread through water. The exact molecule paths vary, but the visible color trend is measurable.
Assessment evidence	Students should connect extraction to diffusion, explain why warmer water usually extracts faster, and distinguish taste preference from measured concentration.
Teacher emphasis	This lab softens the transition from heat to food/body systems. It is chemistry, thermodynamics, and statistical mechanics in a cup.

Differentiation / extension: Support: use visual color rankings. Extend: use image RGB values as a crude concentration proxy.

Lab 8 · Purple Power: Design Your Own Sport Drink

A Gatorade-style design challenge using concentration, sugar energy, salt/electrolytes, taste, and repeatable measurement.



g/mL · kcal · taste

Suggested time	2 class periods or one extended block
Key model	sugar concentration = g sugar/mL drink; kcal from sugar ≈ 4 × g sugar
Materials	Water, sugar, tiny measured salt portions, lemon/lime juice or flavoring, purple food color or grape flavor, cups, scale, measuring spoon, labels, optional pH paper.
Safety	Use food-safe ingredients only; check allergies; no caffeine or supplements; tasting is optional; use clean cups and discard mixtures after lab.

Launch	Show a commercial sports drink label and ask: “What would we have to measure to make a drink with a purpose, not just a color?” Students should name water volume, sugar mass, salt amount, flavor, and serving size.
Investigation	Teams design a Purple Power prototype. They choose volume, sugar mass, a tiny salt/electrolyte amount, and flavor/acid. They calculate sugar concentration and kcal per serving, then revise the recipe for clarity, taste, and repeatability.
Discussion hinge	The hinge is design under constraints. A useful drink is not “more sugar is better.” It is a measured balance among water, energy, electrolytes, flavor, safety, and purpose. This bridges Tea Time chemistry to human work in the step-test labs.
Assessment evidence	Students should submit a repeatable recipe, concentration calculation, kcal estimate, design rationale, and safety statement. Strong work explains tradeoffs and avoids medical claims.
Teacher emphasis	Keep the tone playful but rigorous. “Purple Power” is a formulation lab: students design, measure, calculate, revise, and defend.

Differentiation / extension: Support: give a base recipe with blanks. Extend: optimize a target such as lower sugar, stronger flavor, or best repeatability.

Lab 9 · Harvard Step Test

Mechanical work, power, recovery, and safe personalized effort.



W = mghN

Suggested time	1–2 class periods
Key model	W = mghN; P = W/t
Materials	Low stable step, stopwatch, measuring tape, optional metronome, optional heart-rate monitor, data sheet.
Safety	Participation is voluntary; allow observer/data roles; use low steps; stop for pain, dizziness, shortness of breath, or distress; no ranking bodies.

Launch	Ask: “Is walking up a step the same task for every body?” Then write W = mghN. The formula makes the fairness problem visible without blaming the person.
Investigation	Students use anonymized code/data sets. They measure step height, count steps, record time, and calculate work and power. Optional participants monitor recovery with safe self-paced protocols.
Discussion hinge	The hinge is that the same external task can represent different work for different bodies. Fair measurement requires mass, height, repetitions, time, and recovery context.
Assessment evidence	Students should calculate work and power from coded data, explain why raw speed is not enough, and describe at least one safety/equity rule.
Teacher emphasis	This is the mathematical beginning of Fair Fitness Testing. The goal is not athletic ranking; it is humane energy accounting.

Differentiation / extension: Support: use sample coded data only. Extend: compare power output across step heights while keeping safety rules fixed.

Lab 10 · FFT: Fair Fitness Testing for Populations

Population patterns, anonymized data, variability, fairness, and interpretation.

distribution · fairness

Suggested time	1 class period plus optional project time
Key model	fair comparison = measured work + power + recovery + context
Materials	Anonymized data sets, calculator/spreadsheet, graph paper or dashboard, CER writing sheet.
Safety	Use coded data; no public comparison of bodies; emphasize improvement, safety, and interpretation over ranking.

Launch	Display two anonymized data sets with different masses, step counts, and recovery. Ask: “Who worked harder? What would you need to know before answering fairly?”
Investigation	Students calculate work, power, and recovery indicators for coded data sets. They graph distributions and identify patterns without naming or ranking individuals.
Discussion hinge	The hinge is population reasoning. Individual results vary, but group patterns can reveal whether a test is fair, biased, safe, or useful.
Assessment evidence	Students complete a Claim-Evidence-Reasoning argument that uses at least two measured quantities and includes an equity/safety statement.
Teacher emphasis	Bring back the governing sentence: many small events - steps, heartbeats, recovery intervals, data points - create measurable patterns that can be used responsibly.

Differentiation / extension: Support: use a prepared graph. Extend: propose a revised population test protocol and defend the fairness rule.

Final Synthesis & Extension

The final synthesis should not be a generic reflection. Ask students to complete the arc in one sentence and one chain of reasoning.

Synthesis sentence frame: The same measurement logic that lets us infer hidden size, hidden rate, hidden molecular motion, and hidden energy cost also lets us design fairer ways to compare human work.

RECOMMENDED FINAL PROMPT

Complete this chain using at least one example from each cluster: **measurement → probability/pattern → energy transfer → human application → fairness rule**. Then explain where uncertainty enters the chain and how a responsible scientist reduces harm.

OPTIONAL EXTENSIONS

- Spreadsheet model of RED/BLUE trials as a binomial distribution.
- Newton cooling curve fit and time constant estimate.
- Purple Power optimization challenge with a defined sugar concentration target.
- FFT anonymized data dashboard with histogram and scatterplot views.
- Short research poster: “What makes a fitness test fair?”

Teacher Glossary

Term	Teacher meaning
Accumulation	Total amount gathered over time; often rate × time.
Calorimetry	Measuring heat transfer by observing temperature change.
Entropy	A measure related to how many microscopic arrangements produce the same broad state.
Equilibrium	A shared condition reached when net change becomes very small.
Fair Fitness Testing	A measurement approach that compares work, power, recovery, and context rather than ranking bodies crudely.
Macrostate	A broad visible category, such as “3 red and 2 blue.”
Microstate	One exact arrangement inside a macrostate.
Purple Power	A sport-drink design lab using concentration, recipe constraints, sugar energy, and safety language.
Specific heat	Energy required to raise 1 gram of a material by 1 degree Celsius; for water, $c \approx 4.18$ J/(g·°C).
Work	Energy transferred by a force through a distance; in step testing $W = mghN$.